

P2 — Physical Chemistry II

1. Kinetics (Year 2)

1.1 Rate of Reaction

Rate of reaction: The change in concentration of a reactant or product per unit time. Units: $\text{mol dm}^{-3} \text{ s}^{-1}$.

Rate can be measured as: the gradient of a concentration-time graph at any point (instantaneous rate), or as the initial rate (gradient at time = 0). The initial rate is often preferred because it eliminates complications from reverse reactions and changing concentrations.

1.2 Collision Theory and Activation Energy

For a reaction to occur, particles must collide with sufficient energy (greater than or equal to the activation energy, E_a) and with the correct orientation. Increasing temperature, concentration, pressure, surface area, or using a catalyst all increase the rate by different mechanisms.

Activation energy (E_a): The minimum energy that colliding particles must have for a reaction to occur. It represents the energy barrier between reactants and products on an enthalpy profile diagram.

1.3 Maxwell-Boltzmann Distribution

The Maxwell-Boltzmann distribution shows the spread of molecular energies in a gas at a given temperature. Key features:

- The curve starts at the origin (no molecules have zero energy).
- The peak represents the most probable energy.

- The curve is asymmetric with a long tail to the right (no maximum energy limit).
- The area under the curve represents the total number of molecules and is constant.
- The area to the right of E_a represents molecules with enough energy to react.

Effect of temperature increase: The peak shifts right and becomes lower and broader. More molecules have energy $\geq E_a$, so the rate increases. The total area under the curve remains the same. The key reason for the rate increase is the much larger proportion of molecules exceeding E_a , not simply more frequent collisions.

Effect of a catalyst: Provides an alternative pathway with a lower E_a . The distribution curve stays the same, but the E_a line moves left, so more molecules have sufficient energy. A catalyst does not change the distribution of energies, the position of equilibrium, or ΔH .

1.4 Rate Equations and Orders of Reaction

Order of reaction (with respect to a reactant): The power to which the concentration of that reactant is raised in the rate equation. Orders can only be determined experimentally — they cannot be deduced from the balanced equation.

For a reaction $A + B \rightarrow \text{products}$:

$$\text{Rate} = k[A]^m[B]^n$$

Where k = rate constant, m = order with respect to A, n = order with respect to B. The overall order = $m + n$.

Order	Effect of doubling [reactant]	Graph: [reactant] vs time	Graph: rate vs [reactant]	Half-life
0	No effect on rate	Straight line (linear decrease)	Horizontal line	Varies (decreases over time)
1	Rate doubles	Exponential decay curve	Straight line through origin	Constant

2	Rate quadruples (2^2)	Steeper curve than first order	Upward curve (parabola through origin)	Varies (increases over time)
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Determining orders experimentally — the initial rates method:

1. Carry out several experiments varying one concentration at a time while keeping others constant.
2. Measure the initial rate for each experiment.
3. Compare pairs of experiments where only one concentration changes.
4. If doubling [A] doubles the rate: order 1 with respect to A.
5. If doubling [A] quadruples the rate: order 2 with respect to A.
6. If doubling [A] has no effect on rate: order 0 with respect to A.

Worked Example: Given the following data for $A + B \rightarrow \text{products}$:

Expt	[A] / mol dm ⁻³	[B] / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	0.10	0.10	1.5×10^{-4}
2	0.20	0.10	6.0×10^{-4}
3	0.10	0.20	3.0×10^{-4}

Comparing Expts 1 & 2: [A] doubles, [B] constant, rate quadruples ($6.0/1.5 = 4$). So order 2 w.r.t. A.

Comparing Expts 1 & 3: [A] constant, [B] doubles, rate doubles ($3.0/1.5 = 2$). So order 1 w.r.t. B.

$$\text{Rate} = k[A]^2[B]$$

Finding k: $1.5 \times 10^{-4} = k \times (0.10)^2 \times 0.10 = k \times 0.001$

$$k = 1.5 \times 10^{-4} / 0.001 = \mathbf{0.15 \text{ dm}^6 \text{ mol}^{-2} \text{ s}^{-1}}$$

Exam Tip: The rate equation cannot be deduced from the stoichiometric equation. It must be determined experimentally. The orders tell you about the rate-determining step, not the overall

equation. Always show your reasoning clearly when deducing orders from data.

1.5 The Rate Constant (k) and its Units

Units of k depend on the overall order of the reaction:

Overall order	Units of k	Derivation
0	$\text{mol dm}^{-3} \text{s}^{-1}$	Rate / 1
1	s^{-1}	Rate / [A]
2	$\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$	Rate / [A] ²
3	$\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$	Rate / [A] ³

The value of k increases with temperature because at higher temperature, a greater proportion of collisions are successful. The relationship is quantified by the Arrhenius equation.

1.6 The Arrhenius Equation

$$k = Ae^{-E_a/RT}$$

Where k = rate constant, A = the pre-exponential factor (Arrhenius constant, related to collision frequency and orientation), E_a = activation energy (J mol^{-1}), $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, T = temperature (K).

The exponential term $e^{-E_a/RT}$ represents the fraction of molecules with energy greater than E_a . As T increases, E_a/RT decreases, so $e^{-E_a/RT}$ increases, meaning k increases.

Taking the natural logarithm gives a linear form:

$$\ln k = \ln A - E_a / RT$$

This has the form $y = mx + c$. A graph of $\ln k$ (y-axis) against $1/T$ (x-axis) gives a straight line with:

- Gradient = $-E_a/R$
- y-intercept = $\ln A$

Worked Example: From an Arrhenius plot, the gradient is found to be -1.25×10^4 K. Calculate E_a .

$$\text{Gradient} = -E_a/R$$

$$E_a = -\text{gradient} \times R = -(-1.25 \times 10^4) \times 8.314 = 103,925 \text{ J mol}^{-1} = \mathbf{104 \text{ kJ mol}^{-1}}$$

Exam Tip: To calculate E_a from an Arrhenius plot: $\text{gradient} = -E_a/R$, so $E_a = -\text{gradient} \times R$. Use $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$. The answer will be in J mol^{-1} ; divide by 1000 to convert to kJ mol^{-1} . When calculating the gradient, take two points far apart on the best-fit line (not data points).

1.7 Rate-Determining Step

Rate-determining step (RDS): The slowest step in a multi-step reaction mechanism. It acts as a bottleneck and controls the overall rate of reaction.

The species that appear in the rate equation are those that appear in or before the rate-determining step. The molecularity of the RDS corresponds to the rate equation:

- If $\text{Rate} = k[A][B]$, the RDS involves one molecule of A and one molecule of B (bimolecular).
- If $\text{Rate} = k[A]^2$, the RDS involves two molecules of A.
- If $\text{Rate} = k[A]$ (and B is also a reactant), then B is not involved in or before the RDS — it must be involved in a fast step after the RDS.

Example — SN1 reaction: The hydrolysis of tertiary halogenoalkanes has $\text{Rate} = k[\text{RX}]$. This tells us the RDS only involves the halogenoalkane, consistent with the slow heterolytic fission of the C-X bond (step 1), followed by fast nucleophilic attack (step 2).

Exam Tip: If asked to propose a mechanism consistent with a rate equation, check that the species and their stoichiometry in the RDS match the rate equation exactly. Species absent from the rate equation must appear in steps after the RDS.

2. Equilibria

2.1 Dynamic Equilibrium

Dynamic equilibrium: A state reached in a closed system when the rate of the forward reaction equals the rate of the reverse reaction, so the concentrations of reactants and products remain constant (though not necessarily equal). Both forward and reverse reactions continue to occur — the equilibrium is dynamic, not static.

Conditions for dynamic equilibrium:

- A closed system (nothing enters or leaves).
- A reversible reaction.
- The forward and reverse rates are equal.
- Macroscopic properties (concentrations, colour, pressure) remain constant.

2.2 K_c — The Equilibrium Constant (Concentrations)

For the reaction: $aA + bB \rightleftharpoons cC + dD$

$$K_c = [C]^c[D]^d / [A]^a[B]^b$$

Where square brackets denote equilibrium concentrations in mol dm^{-3} . K_c has units that must be worked out from the expression (they depend on the powers).

Interpreting K_c :

- $K_c \gg 1$: Equilibrium lies far to the right — products strongly favoured.
- $K_c \ll 1$: Equilibrium lies far to the left — reactants strongly favoured.
- $K_c \approx 1$: Significant amounts of both reactants and products present.

Worked Example: For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, at equilibrium $[\text{N}_2\text{O}_4] = 0.0400 \text{ mol dm}^{-3}$ and $[\text{NO}_2] = 0.0280 \text{ mol dm}^{-3}$. Calculate K_c .

$$K_c = [\text{NO}_2]^2 / [\text{N}_2\text{O}_4] = (0.0280)^2 / 0.0400 = 7.84 \times 10^{-4} / 0.0400 = \mathbf{0.0196 \text{ mol dm}^{-3}}$$

$$\text{Units: } (\text{mol dm}^{-3})^2 / (\text{mol dm}^{-3}) = \text{mol dm}^{-3}$$

2.3 K_p — Equilibrium Constant (Partial Pressures)

For gaseous equilibria, we can use partial pressures instead of concentrations:

Mole fraction: $x_A = \text{moles of A} / \text{total moles of gas}$

Partial pressure: $p_A = x_A \times P_{\text{total}}$

$$K_p = p_C^c \times p_D^d / (p_A^a \times p_B^b)$$

Worked Example: In the Haber process $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, at equilibrium the mole fractions are: $x(\text{N}_2) = 0.20$, $x(\text{H}_2) = 0.60$, $x(\text{NH}_3) = 0.20$. Total pressure = 200 atm. Calculate K_p .

$$p(\text{N}_2) = 0.20 \times 200 = 40 \text{ atm}$$

$$p(\text{H}_2) = 0.60 \times 200 = 120 \text{ atm}$$

$$p(\text{NH}_3) = 0.20 \times 200 = 40 \text{ atm}$$

$$K_p = (40)^2 / [(40) \times (120)^3] = 1600 / (40 \times 1,728,000) = 1600 / 69,120,000 = 2.31 \times 10^{-5} \text{ atm}^{-2}$$

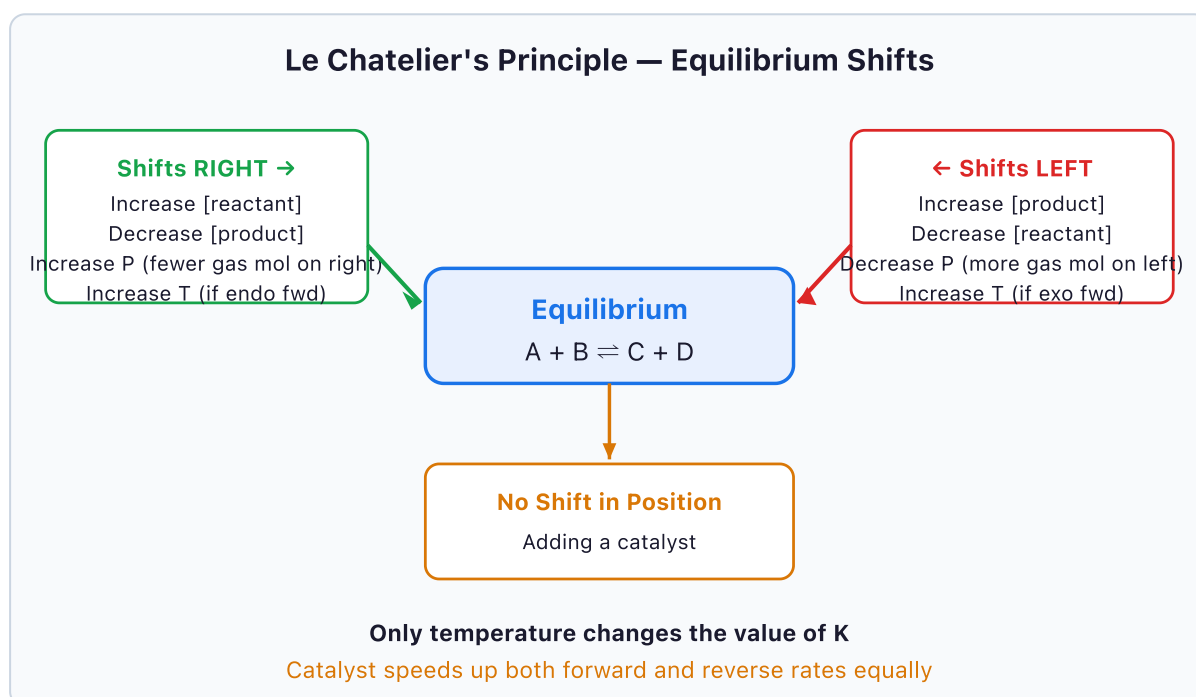
2.4 Le Chatelier's Principle

Le Chatelier's Principle: If a system at dynamic equilibrium is subjected to a change in conditions (concentration, temperature, or pressure), the position of equilibrium shifts in the direction that tends to counteract the change and restore equilibrium.

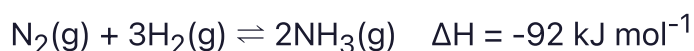
Change	Effect on position of equilibrium	Effect on K
Increase concentration of reactant	Shifts right (towards products)	No change
Decrease concentration of product	Shifts right (towards products)	No change
Increase temperature (exothermic forward)	Shifts left (endothermic direction)	K decreases

Increase temperature (endothermic forward)	Shifts right (endothermic direction)	K increases
Increase pressure	Shifts to side with fewer moles of gas	No change
Add catalyst	No change in position	No change

Critical point: Only temperature changes the value of K_c or K_p . Changes in concentration, pressure, or catalysts change the position of equilibrium but do NOT change K. A catalyst speeds up both forward and reverse reactions equally, so equilibrium is reached faster but at the same position.



2.5 Industrial Application — The Haber Process



This is an exothermic reaction with fewer moles of gas on the right (2 vs 4). Applying Le Chatelier's principle:

- **Low temperature** favours the exothermic forward reaction (higher yield) but makes the reaction too slow. Compromise: **450 °C**.
- **High pressure** favours the side with fewer moles of gas (products). Compromise: **200 atm** (higher pressure is more expensive and dangerous).

- **Iron catalyst** increases the rate of reaching equilibrium (does not change yield) — essential for economic viability.
- Ammonia is continuously removed by cooling and liquefying — this shifts the equilibrium to the right, increasing overall conversion.

Exam Tip: A common exam error is saying that a catalyst increases the yield. It does NOT. A catalyst only increases the rate at which equilibrium is reached. The yield (position of equilibrium) is determined only by temperature and pressure.

3. Oxidation, Reduction and Redox

3.1 Oxidation States

Oxidation state (oxidation number): A measure of the degree of oxidation of an atom in a compound. It represents the charge an atom would have if all bonds were completely ionic.

Rules for assigning oxidation states:

- Uncombined elements = 0 (e.g. Na, O₂, S₈)
- Simple ions = the charge on the ion (e.g. Na⁺ = +1, Cl⁻ = -1, Fe³⁺ = +3)
- Fluorine = always -1 (most electronegative element)
- Oxygen = -2 (except in peroxides O₂²⁻ = -1, and OF₂ = +2)
- Hydrogen = +1 (except in metal hydrides like NaH = -1)
- Sum of oxidation states in a neutral compound = 0
- Sum of oxidation states in a polyatomic ion = the overall charge on the ion

Oxidation: Loss of electrons / increase in oxidation state. (OIL)

Reduction: Gain of electrons / decrease in oxidation state. (RIG)

Oxidising agent: A substance that oxidises another and is itself reduced (gains electrons).

Reducing agent: A substance that reduces another and is itself oxidised (loses electrons).

Worked Example: Find the oxidation state of Mn in KMnO_4 .

K = +1 (Group 1 metal), each O = -2, let Mn = x.

$$(+1) + x + 4(-2) = 0$$

$$+1 + x - 8 = 0$$

$$x = +7$$

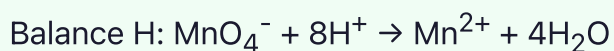
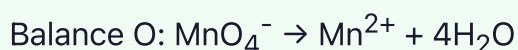
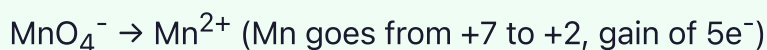
3.2 Half-Equations and Ionic Equations

Redox reactions can be split into two half-equations — one for oxidation and one for reduction. To balance half-equations in acidic solution:

1. Balance the atoms other than O and H.
2. Balance oxygen by adding H_2O molecules.
3. Balance hydrogen by adding H^+ ions.
4. Balance charge by adding electrons (e^-).

To combine half-equations: multiply so the electron numbers are equal, then add and cancel.

Worked Example: Balance the half-equation for the reduction of MnO_4^- to Mn^{2+} in acidic solution.



Balance charge: left = $-1 + 8 = +7$; right = $+2$. Need 5e^- on left.



3.3 Redox Titrations

Potassium manganate(VII) (KMnO_4) is commonly used as an oxidising agent in titrations. It is self-indicating: purple MnO_4^- is decolourised as it reacts, turning to almost colourless Mn^{2+} . The end point is reached when a permanent pink/purple colour persists.

Common titrations involve manganate(VII) with iron(II) or with ethanedioic acid (oxalic acid):



4. Thermodynamics

4.1 Born-Haber Cycles

A Born-Haber cycle applies Hess's law to calculate lattice enthalpy indirectly. The cycle links several enthalpy terms:

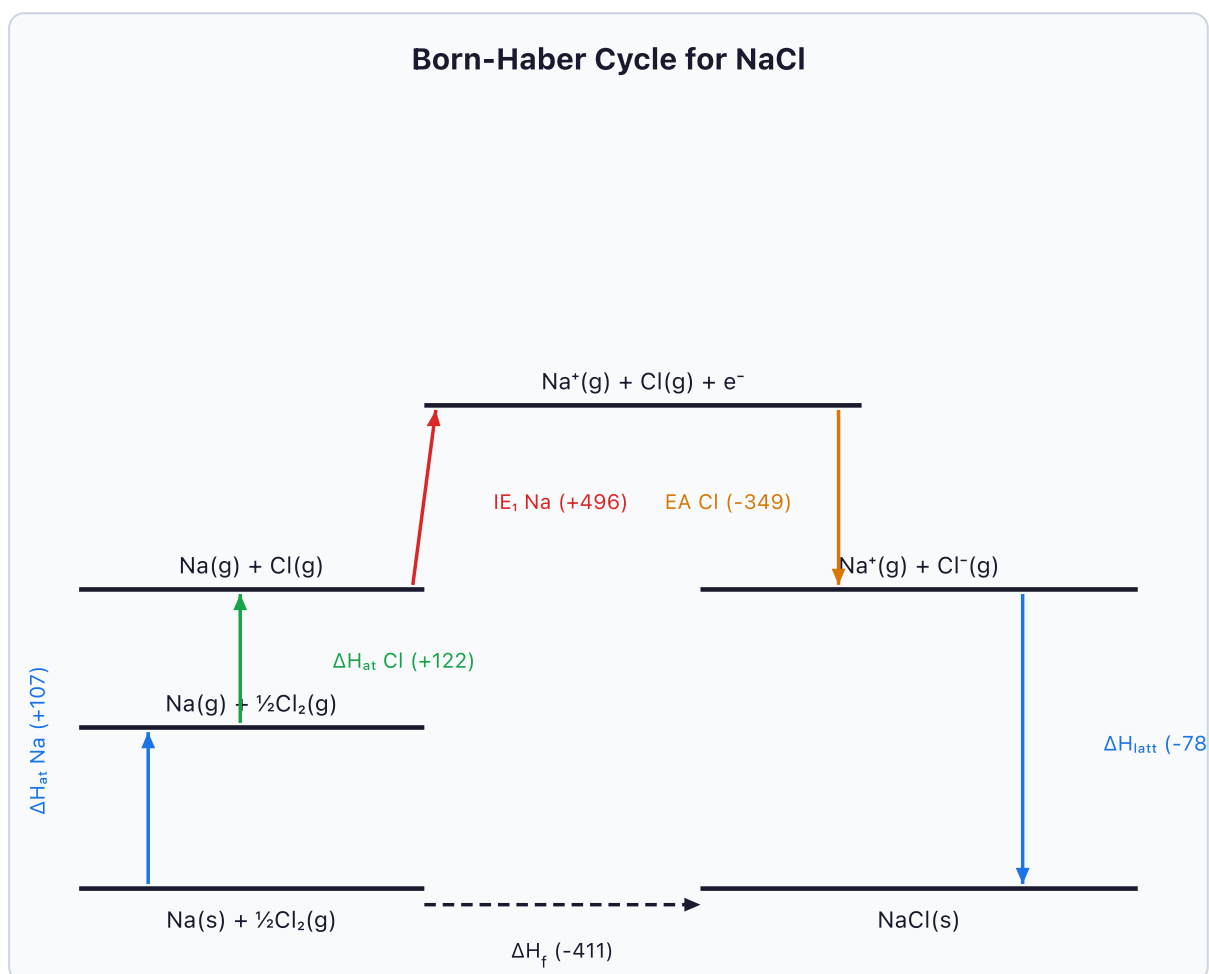
Term	Definition	Sign
ΔH_f°	Standard enthalpy of formation of the ionic compound from its elements	Usually negative
$\Delta H_{\text{at}}^\circ$	Enthalpy of atomisation — forming one mole of gaseous atoms from the element	Always positive
IE	Ionisation energy — removing electrons from gaseous atoms to form cations	Always positive
EA	Electron affinity — gaseous atom gaining an electron to form an anion	1st EA usually negative; 2nd EA positive
ΔH_{latt}	Lattice enthalpy — see definition below	Depends on convention

Lattice enthalpy of formation (ΔH_{latt}): The enthalpy change when one mole of an ionic lattice is formed from its gaseous ions under standard conditions. Always exothermic (negative). This is the convention used in Born-Haber cycles on AQA.

Lattice enthalpy of dissociation: The enthalpy change when one mole of an ionic lattice is completely separated into gaseous ions. Always endothermic (positive). This is the reverse of the formation definition.

Factors affecting lattice enthalpy magnitude:

- Ionic charge:** Higher charges = stronger electrostatic attraction = more exothermic lattice enthalpy. MgO has a much larger lattice enthalpy than NaCl because Mg^{2+} and O^{2-} have double the charges of Na^+ and Cl^- .
- Ionic radius:** Smaller ions = ions closer together = stronger attraction = more exothermic. LiF has a larger lattice enthalpy than CsI because Li^+ and F^- are much smaller than Cs^+ and I^- .



Theoretical vs experimental lattice enthalpies: Theoretical values are calculated assuming a purely ionic model (spherical ions with no distortion). If the experimental value (from Born-Haber) is more exothermic than the theoretical value, this indicates significant covalent character in the bonding — the ions are polarised. This is common when cations are small and highly charged (high charge density), or anions are large and easily polarised (e.g. I^-).

4.2 Enthalpy of Solution and Hydration

Enthalpy of hydration (ΔH_{hyd}): The enthalpy change when one mole of gaseous ions is dissolved in sufficient water to form an infinitely dilute solution. Always exothermic because ion-dipole bonds form between the ion and water molecules.

Enthalpy of solution (ΔH_{sol}): The enthalpy change when one mole of a solute dissolves in sufficient water to form an infinitely dilute solution.

$$\Delta H_{\text{sol}} = -\Delta H_{\text{latt}}(\text{formation}) + \Delta H_{\text{hyd}}(\text{cation}) + \Delta H_{\text{hyd}}(\text{anion})$$

Or equivalently: $\Delta H_{\text{sol}} = \Delta H_{\text{latt}}(\text{dissociation}) + \Delta H_{\text{hyd}}(\text{cation}) + \Delta H_{\text{hyd}}(\text{anion})$

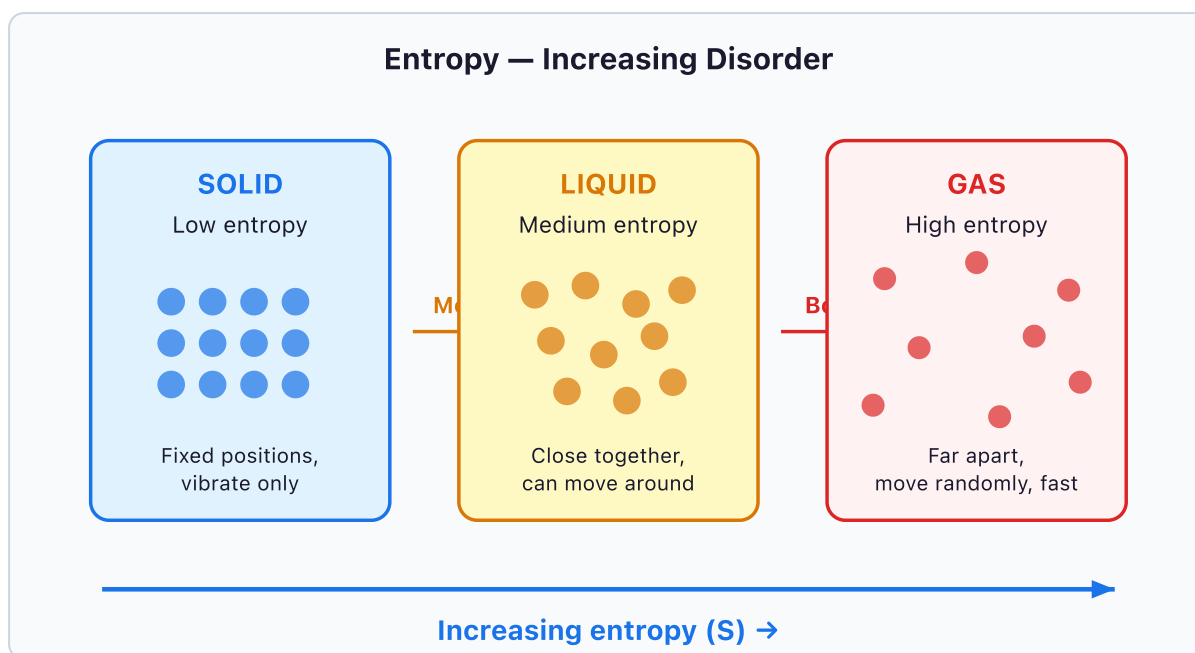
A substance dissolves if ΔH_{sol} is exothermic (or only slightly endothermic, as entropy increase can compensate). If the lattice enthalpy is too large relative to the hydration enthalpies, the substance is insoluble.

4.3 Entropy

Entropy (S): A measure of the disorder or dispersal of energy in a system — the number of possible arrangements of particles and energy. Units: $\text{J K}^{-1} \text{mol}^{-1}$. Higher entropy means more disorder.

Entropy increases when:

- Solids melt to liquids (particles gain freedom of movement)
- Liquids evaporate to gases (massive increase in disorder)
- The number of moles of gas increases (more gaseous particles = more ways to distribute energy)
- A solid dissolves in a solvent (particles become dispersed)
- Temperature increases (more energy quanta to distribute)



$$\Delta S_{\text{system}} = \sum S^{\circ}(\text{products}) - \sum S^{\circ}(\text{reactants})$$

Worked Example: Calculate ΔS for $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$. Given: $S^{\circ}[\text{CaCO}_3] = 93$, $S^{\circ}[\text{CaO}] = 40$, $S^{\circ}[\text{CO}_2] = 214 \text{ J K}^{-1} \text{ mol}^{-1}$.

$$\Delta S = (40 + 214) - 93 = \mathbf{+161 \text{ J K}^{-1} \text{ mol}^{-1}}$$

This is positive because a gas is produced (1 mole of solid $\rightarrow$ 1 mole of solid + 1 mole of gas).

4.4 Gibbs Free Energy

$$\Delta G = \Delta H - T\Delta S$$

A reaction is thermodynamically feasible (spontaneous) when $\Delta G \leq 0$.

ΔH	ΔS	Feasibility	Example
Negative (exo)	Positive (+ ΔS)	Always feasible (ΔG always negative)	Combustion of methane

Negative (exo)	Negative ($-\Delta S$)	Feasible at low temperatures only	Freezing of water
Positive (endo)	Positive ($+\Delta S$)	Feasible at high temperatures only	Thermal decomposition of CaCO_3
Positive (endo)	Negative ($-\Delta S$)	Never feasible (ΔG always positive)	Very rare in practice

The temperature at which an endothermic, positive- ΔS reaction becomes feasible can be found by setting $\Delta G = 0$:

$$T = \Delta H / \Delta S \quad (\text{at the boundary of feasibility})$$

Worked Example: For $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, $\Delta H = +178 \text{ kJ mol}^{-1}$ and $\Delta S = +161 \text{ J K}^{-1} \text{ mol}^{-1}$. Find the temperature above which the reaction is feasible.

Convert units: $\Delta S = 0.161 \text{ kJ K}^{-1} \text{ mol}^{-1}$

$$T = \Delta H / \Delta S = 178 / 0.161 = \mathbf{1106 \text{ K (833 } ^\circ\text{C)}}$$

Above this temperature, the $T\Delta S$ term outweighs ΔH , making ΔG negative.

Exam Tip: When using $\Delta G = \Delta H - T\Delta S$, ensure units are consistent. ΔH is usually in kJ mol^{-1} and ΔS in $\text{J K}^{-1} \text{ mol}^{-1}$. Convert ΔS to kJ (divide by 1000) or ΔH to J (multiply by 1000) before calculating. A reaction with $\Delta G \leq 0$ is feasible but may not occur in practice if the activation energy is too high (kinetic barrier).

5. Acids and Bases

5.1 Bronsted-Lowry Theory

Bronsted-Lowry acid: A proton (H^+) donor.

Bronsted-Lowry base: A proton (H^+) acceptor.

Conjugate acid-base pairs: When an acid donates a proton, it forms its conjugate base. When a base accepts a proton, it forms its conjugate acid. For example:



$\text{CH}_3\text{COOH}/\text{CH}_3\text{COO}^-$ is one conjugate pair; $\text{H}_2\text{O}/\text{H}_3\text{O}^+$ is the other.

Strong acids (HCl , H_2SO_4 , HNO_3) fully dissociate in water — the equilibrium lies completely to the right.

Weak acids (CH_3COOH , H_2CO_3 , HF) only partially dissociate — an equilibrium is established, with most molecules remaining undissociated.

5.2 pH and $[\text{H}^+]$ Calculations

$$\text{pH} = -\log_{10}[\text{H}^+] \quad \text{and} \quad [\text{H}^+] = 10^{-\text{pH}}$$

Strong acid pH: For a monobasic strong acid (e.g. HCl): $[\text{H}^+] = \text{concentration of acid}$. For dibasic H_2SO_4 : $[\text{H}^+] = 2 \times \text{concentration}$.

Strong base pH: Calculate $[\text{OH}^-]$ first, then use K_w :

$$K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6} \text{ at } 298 \text{ K}$$

At 298 K: $[\text{H}^+] = K_w / [\text{OH}^-]$, then $\text{pH} = -\log[\text{H}^+]$.

K_w increases with temperature because the dissociation of water is endothermic ($\Delta H = +57.1 \text{ kJ mol}^{-1}$).

At higher temperatures, $[\text{H}^+]$ and $[\text{OH}^-]$ both increase (K_w is larger), so pH of pure water decreases below 7 — but the water is still neutral because $[\text{H}^+] = [\text{OH}^-]$.

Worked Example: Calculate the pH of $0.050 \text{ mol dm}^{-3} \text{ NaOH}$ at 298 K.

$[\text{OH}^-] = 0.050 \text{ mol dm}^{-3}$ (strong base, fully dissociates)

$$[\text{H}^+] = K_w / [\text{OH}^-] = 1.00 \times 10^{-14} / 0.050 = 2.00 \times 10^{-13} \text{ mol dm}^{-3}$$

$$\text{pH} = -\log(2.00 \times 10^{-13}) = \mathbf{12.70}$$

5.3 The Acid Dissociation Constant K_a

For a weak acid HA:



$$K_a = [\text{H}^+][\text{A}^-] / [\text{HA}]$$

Assumptions for weak acid pH calculations:

- $[\text{H}^+] = [\text{A}^-]$ (because virtually all H^+ comes from the weak acid, not from water).
- $[\text{HA}]_{\text{equilibrium}} \approx [\text{HA}]_{\text{initial}}$ (very little of the acid has dissociated).

Therefore:

$$K_a \approx [\text{H}^+]^2 / [\text{HA}]_{\text{initial}} \quad \text{so} \quad [\text{H}^+] = \sqrt{(K_a \times [\text{HA}])}$$

$$\text{p}K_a = -\log_{10} K_a$$

A smaller K_a (larger $\text{p}K_a$) means a weaker acid — less dissociation.

Worked Example: Calculate the pH of $0.100 \text{ mol dm}^{-3}$ ethanoic acid ($K_a = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$).

$$[\text{H}^+] = \sqrt{(K_a \times [\text{HA}])} = \sqrt{(1.74 \times 10^{-5} \times 0.100)} = \sqrt{(1.74 \times 10^{-6})} = 1.32 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\text{pH} = -\log(1.32 \times 10^{-3}) = \mathbf{2.88}$$

5.4 Buffer Solutions

Buffer solution: A solution that resists changes in pH when small amounts of acid or base are added, or when diluted with water. It consists of a weak acid and its conjugate base (or a weak base and its conjugate acid) in roughly equal concentrations.

Making an acidic buffer:

- Mix a weak acid with its sodium/potassium salt (e.g. $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$).
- Or partially neutralise a weak acid with a strong base (e.g. add NaOH to excess CH_3COOH).

How a buffer works (using $\text{CH}_3\text{COOH} / \text{CH}_3\text{COO}^-$ as example):

- **If H^+ is added:** The conjugate base (CH_3COO^- , present in large reserve) reacts with the added H^+ :
 $\text{CH}_3\text{COO}^- + \text{H}^+ \rightarrow \text{CH}_3\text{COOH}$. This removes the added H^+ , preventing pH from falling significantly.
- **If OH^- is added:** The weak acid (CH_3COOH , present in large reserve) reacts with the added OH^- :
 $\text{CH}_3\text{COOH} + \text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$. This removes the added OH^- , preventing pH from rising significantly.

Calculating buffer pH:

$$[\text{H}^+] = K_a \times [\text{HA}] / [\text{A}^-]$$

Since $[\text{HA}]$ and $[\text{A}^-]$ are in the same volume of solution, you can use moles instead of concentrations.

Worked Example: A buffer is made by mixing 100 cm^3 of $0.200 \text{ mol dm}^{-3}$ CH_3COOH with 50 cm^3 of $0.200 \text{ mol dm}^{-3}$ NaOH. Calculate its pH. ($K_a = 1.74 \times 10^{-5}$)

Moles $\text{CH}_3\text{COOH} = 0.100 \times 0.200 = 0.0200 \text{ mol}$

Moles NaOH = $0.050 \times 0.200 = 0.0100 \text{ mol}$

NaOH neutralises some acid: $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

Moles CH_3COOH remaining = $0.0200 - 0.0100 = 0.0100 \text{ mol}$

Moles CH_3COO^- formed = 0.0100 mol

$$[\text{H}^+] = K_a \times [\text{HA}] / [\text{A}^-] = 1.74 \times 10^{-5} \times (0.0100 / 0.0100) = 1.74 \times 10^{-5}$$

$$\text{pH} = -\log(1.74 \times 10^{-5}) = \mathbf{4.76} \quad (= \text{p}K_a, \text{ because } [\text{HA}] = [\text{A}^-])$$

Biological importance of buffers: Blood is maintained at pH 7.4 by the carbonic acid/hydrogencarbonate buffer system ($\text{H}_2\text{CO}_3/\text{HCO}_3^-$). Enzymes require specific pH ranges to function — even small changes can denature proteins. Buffer systems in the body prevent dangerous pH fluctuations.

Exam Tip: When explaining how a buffer works in exam questions, you **MUST** name the specific species that react (e.g. "the ethanoate ions react with the added H^+ " not just "the conjugate base"). Include the equation for each case (adding acid and adding base). Also state that the large reserves of both HA and A^- are barely changed by the small addition.

5.5 Titration Curves and Indicators

The shape of a pH titration curve depends on the strengths of the acid and base used:

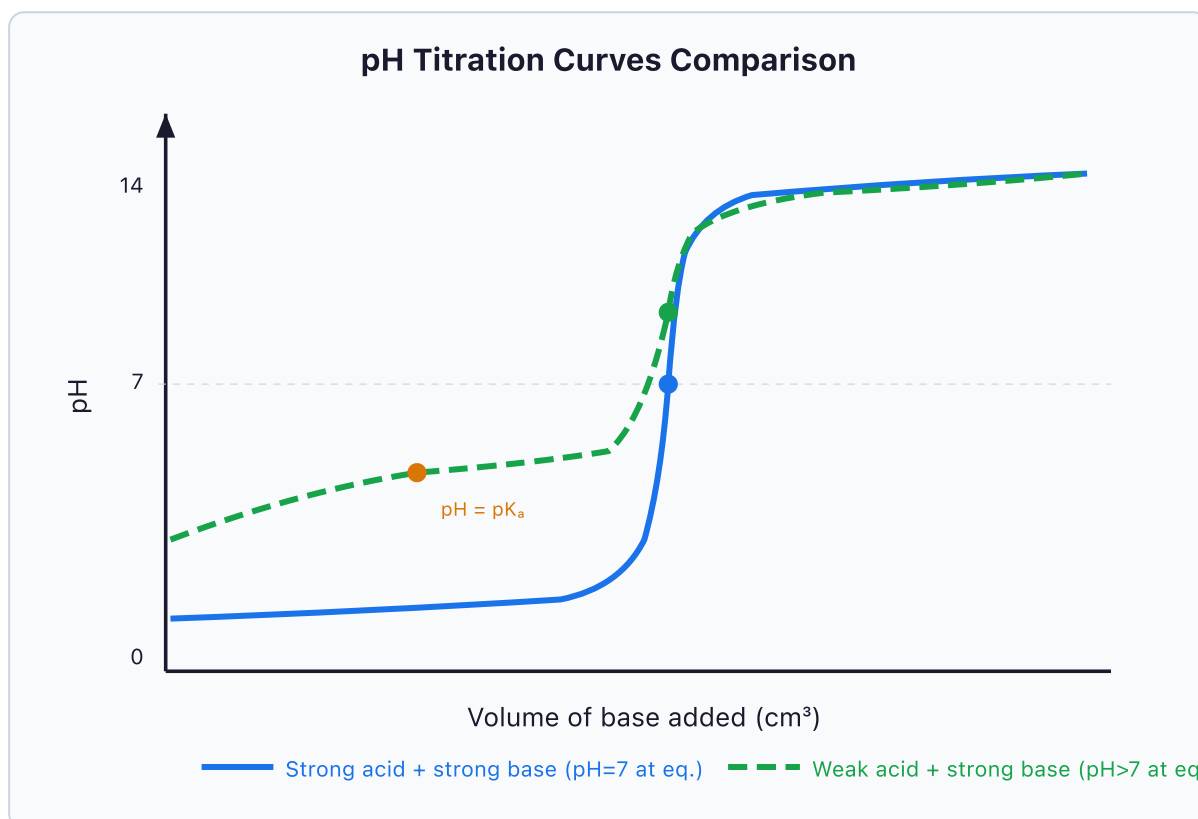
Combination	pH at equivalence point	Suitable indicator	Buffer region?
Strong acid + strong base	7	Any (phenolphthalein or methyl orange)	No
Strong acid + weak base	< 7 (acidic salt formed)	Methyl orange (range 3.1–4.4)	After equivalence
Weak acid + strong base	> 7 (basic salt formed)	Phenolphthalein (range 8.3–10)	Before equivalence
Weak acid + weak base	~7 (gradual change, no sharp endpoint)	No suitable indicator	Both sides

Key features of a weak acid / strong base titration curve:

- Initial pH is higher than a strong acid (weak acid only partially dissociates).
- Buffer region before the equivalence point (mixture of weak acid and its conjugate base).

- Half-equivalence point: $\text{pH} = \text{pK}_a$ (because $[\text{HA}] = [\text{A}^-]$, so $K_a = [\text{H}^+]$).
- Equivalence point: $\text{pH} > 7$ (the salt formed is basic).
- Use phenolphthalein as indicator (its colour change range includes the steep section of the curve).

An indicator changes colour over a range of about 2 pH units. It must be chosen so that its colour change range falls within the steep (vertical) section of the titration curve, ensuring a sharp, precise endpoint.



6. Electrode Potentials and Electrochemical Cells

6.1 Standard Electrode Potential

Standard electrode potential ($E^\ominus$): The EMF (electromotive force) of a half-cell connected to a standard hydrogen electrode under standard conditions: 298 K, 100 kPa, 1.00 mol dm⁻³ solutions of all aqueous species.

The **standard hydrogen electrode (SHE)** is assigned a potential of exactly 0.00 V. It consists of hydrogen gas at 100 kPa bubbled over a platinum electrode in 1.00 mol dm⁻³ H⁺(aq). The half-equation is: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$ $E^\circ = 0.00 \text{ V}$.

More negative E° = stronger reducing agent (greater tendency to lose electrons / be oxidised).

More positive E° = stronger oxidising agent (greater tendency to gain electrons / be reduced).

Common standard electrode potentials:

Half-equation	E° / V
$\text{Li}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Li}(\text{s})$	-3.04
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Zn}(\text{s})$	-0.76
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	-0.44
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$	0.00
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s})$	+0.34
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightleftharpoons \text{Ag}(\text{s})$	+0.80
$\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightleftharpoons \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$	+1.51

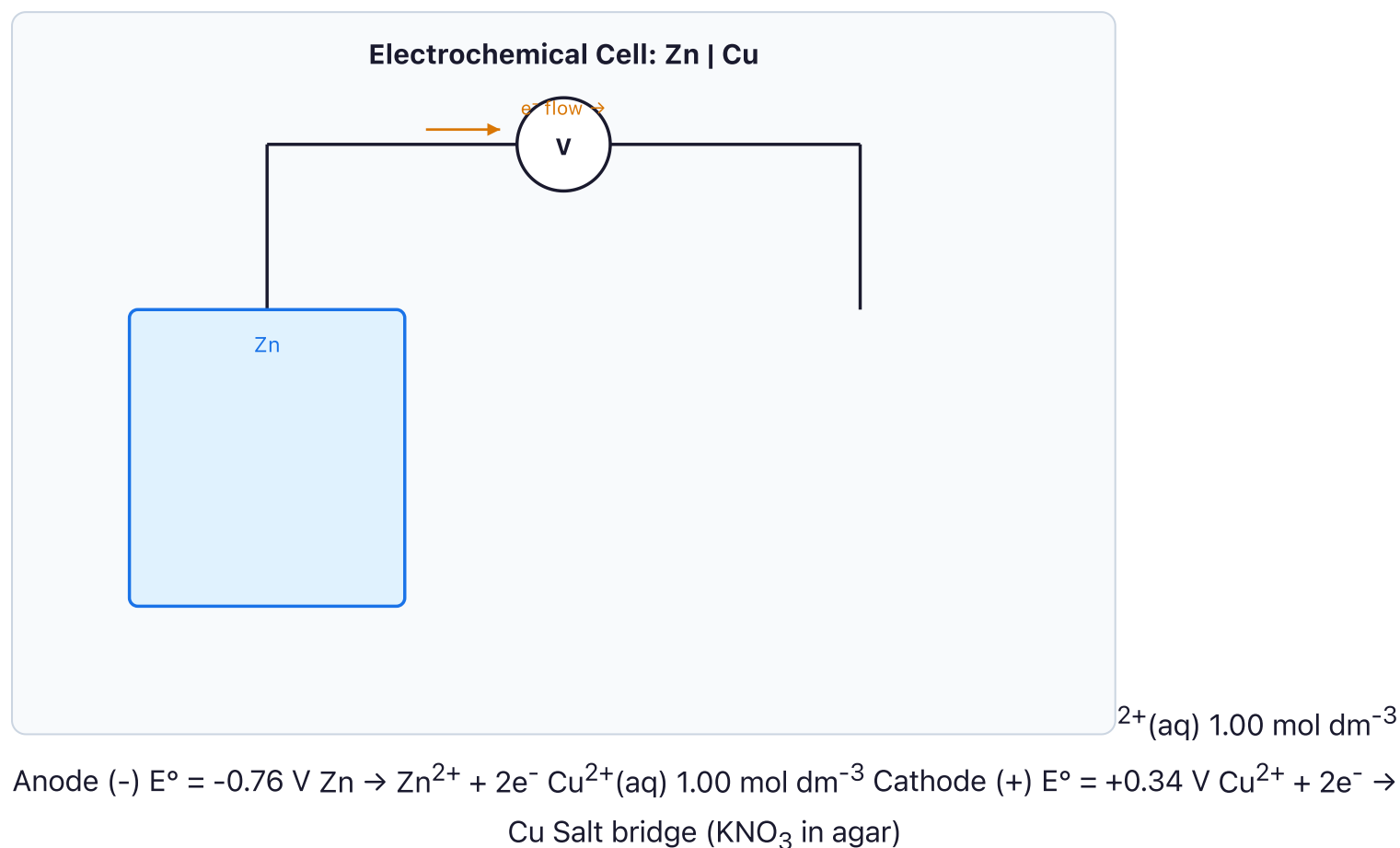
6.2 Electrochemical Cells

An electrochemical cell consists of two half-cells connected by a wire (external circuit) and a salt bridge. The cell EMF is calculated as:

$$E_{\text{cell}} = E^\circ(\text{cathode, more positive}) - E^\circ(\text{anode, more negative})$$

The more positive half-cell is the **cathode** (positive terminal, where reduction occurs). The more negative half-cell is the **anode** (negative terminal, where oxidation occurs). Electrons flow from the anode to the cathode through the external circuit.

The **salt bridge** completes the circuit by allowing ions to flow between the half-cells, maintaining electrical neutrality. It is typically a strip of filter paper soaked in a saturated solution of KNO_3 (which doesn't react with the half-cell solutions).



Worked Example: Calculate E_{cell} for the Zn/Cu cell above.

$$E_{\text{cell}} = E^\circ(\text{cathode}) - E^\circ(\text{anode}) = (+0.34) - (-0.76) = \mathbf{+1.10 \text{ V}}$$

6.3 Feasibility of Reactions from Electrode Potentials

A reaction is thermodynamically feasible if E_{cell} is positive. To predict whether a redox reaction will occur:

1. Identify the two relevant half-equations from the data.
2. The half-cell with the more positive E° undergoes reduction (the forward reaction).
3. The half-cell with the more negative E° undergoes oxidation (the reverse reaction).

4. Calculate $E_{\text{cell}} = E^{\circ}(\text{reduction}) - E^{\circ}(\text{oxidation})$. If positive, the reaction is feasible.

Limitations of electrode potential predictions:

- A positive E_{cell} only indicates **thermodynamic feasibility**. The reaction may be kinetically too slow (high activation energy barrier) — e.g. the reaction of permanganate with oxalic acid is very slow at room temperature despite having a large positive E_{cell} .
- Standard conditions may not apply — non-standard concentrations, temperatures, or pressures can shift electrode potentials.
- The prediction says nothing about the rate of the reaction.

6.4 Storage Cells and Fuel Cells

Non-rechargeable cells (primary cells): The chemical reactions are irreversible (e.g. zinc-carbon batteries).

Rechargeable cells (secondary cells): The reactions are reversible; applying an external voltage reverses the cell reaction (e.g. lithium-ion batteries in phones and laptops, lead-acid car batteries).

Hydrogen fuel cells: Hydrogen and oxygen react electrochemically to produce water and electricity. The overall reaction is $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$. Advantages: high efficiency (no Carnot limit), only product is water (zero emissions at point of use), renewable if hydrogen is produced from electrolysis using renewable energy. Disadvantages: hydrogen storage is difficult (low density, requires high pressure or cryogenic temperatures), hydrogen is flammable, platinum catalysts are expensive, and most hydrogen is currently produced from fossil fuels (steam reforming).

$$\Delta G = -nFE_{\text{cell}}$$

Where n = number of moles of electrons transferred per mole of reaction, F = Faraday constant ($96,485 \text{ C mol}^{-1}$). This equation links electrode potentials to Gibbs free energy. If E_{cell} is positive, ΔG is negative — the reaction is spontaneous.

Exam Tip: When predicting feasibility from E° values, always state three things: (1) E_{cell} is positive so the reaction is thermodynamically feasible; (2) note that this doesn't guarantee the reaction will occur (kinetic factors); (3) the prediction applies under standard conditions only. For fuel cell questions, compare the advantages and disadvantages systematically.

